

HidAttack: An Effective and Undetectable Model Poisoning Attack to Federated Recommenders

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Abstract—Privacy concerns in recommender systems are potentially addressed due to constitutional and commercial requirements. Centralized recommendation models are susceptible to poisoning attacks, which threaten their integrity. In this context, federated learning has emerged as an optimal solution to privacy concerns. However, recent investigations proved that Federated Recommender Systems (FedRS) are also vulnerable to model poisoning attacks. Existing attack possibilities highlighted in academic literature require a large fraction of Byzantine clients to effectively influence the training process, which is unrealistic for practical systems with millions of users. Additionally, most attack models neglected the role of the defense mechanism running at the aggregation server. To this end, we propose a novel undetectable hidden attack strategy (HidAttack) for FedRS, aiming to raise the exposure ratio of targeted items with minimum Byzantine clients. To achieve this goal, we construct a cluster of baseline attacks, on top of which a bandit model is designed that intelligently infers effective poisoned gradients. It ensures a diverse pattern of poisoned gradients and therefore, Byzantine clients cannot be distinguished from benign clients by the defense mechanism. Extensive experiments demonstrate that: 1) our attack model significantly increases the target item's exposure rate covertly without compromising the recommendation accuracy and 2) the current defenses are insufficient, emphasizing the need for better security improvements against our model poisoning attack to FedRS.

Index Terms—Federated learning, model poisoning attacks, privacy protection, responsible recommender systems.

I. INTRODUCTION

RECOMMENDER systems (RS) play an important role in connecting users to interesting items and alleviating the information overload issue. With growing digitalization, the

demand for personalized recommendation services is increasing rapidly. Over the years, diverse recommendation models have been flourished and proven successful. Most conventional recommenders are trained on centralized user data, which has the risk of data leakage and raises privacy concerns [1], [2]. Moreover, recent privacy regulations make it difficult to collect user data for centralized model training [2]. Thus, ensuring the privacy and security of user data has become a critical challenge in RS. Recently, Federated Learning (FL) has emerged as a promising solution to several privacy issues in RS, offering advantages such as utilizing edge resources, decentralized training, and effective personalization.

Recent studies [3], [4], [5], [6] witness that FL models alone are susceptible to poisoning attacks. The decentralized training process allows malicious actors to manipulate local training for desired objectives, e.g., promoting targeted items (targeted attacks) or reducing overall recommendation performance (untargeted attacks) [4], [7], which is usually carried out for financial incentives but incurs strong unfairness in a trained RS [8]. In production, most attack strategies rely on injecting fabricated interactions into the training data [8]. However, due to the limited number of artificial interactions that attackers can introduce, these attacks often exhibit reduced efficacy. In Federated Recommender Systems (FedRS), a key barrier is the limited access to authentic interactions. Conversely, model poisoning attacks, which constitute another category, directly corrupt the model by generating poisoned gradients [4]. These attacks, highly exploitable within FL, frequently surpass the effectiveness of data poisoning attacks. It is important to note that while model poisoning attacks demonstrate greater potential, they require more information about the system, which might not always be feasible for attackers.

Various attack vectors have been explored within the broader field of FL [9], and more specifically within FedRS [5], [10]. While basic approaches like noise injection through fake users or the integration of poisoned data have been mitigated by anomaly detection and intelligent aggregation methods, more sophisticated strategies, such as PipAttack [10] and covert attacks [1], exploit inherent system vulnerabilities to manipulate FedRS for targeted item promotion and performance degradation. However, these methods fail to address several practical considerations:

- 1) These attacks [11], [12] assume the absence of a server-based defense mechanism capable of detecting artificially injected clients. Such an assumption is impractical in production [13].

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- 2) Existing attack strategies [10], [14] necessitate a substantial fraction of Byzantine clients (e.g., 10% - 40%) to significantly impact the training process. This requirement proves unrealistic in practical applications.
- 3) Most attack models [4], [15] rely on the attacker's prior knowledge. This dependency often leads to notable performance degradation within FedRS, making the attack detectable due to its reliance on existing knowledge.

A limited number of studies have addressed the aforementioned limitations in FedRS attacks. For instance, Rong et al. [11], [12] proposed attack models effective under restricted malicious profiles, yet they overlooked the defense mechanism at the aggregation server. Similarly, Zhang et al. [10] introduced an impactful attack strategy, but their model's reliance on a substantial fraction of malicious users impedes its practicality. Moreover, conventional FedRS attacks are largely assumption-based and capability-specific. To tackle these challenges, we propose a novel and undetectable poisoning strategy to promote targeted items within FedRS, employing a limited number of malicious users while effectively evading defense mechanisms. Our strategy draws inspiration from the effectiveness of multi-armed bandit models in generating diverse attack patterns. To construct a robust attack approach, we have amalgamated four established baseline attack models [10], [12], [16], [17] specifically tailored for item promotion within federated settings. The rationale behind integrating these four baseline attack models lies in their distinct strengths and characteristics. However, other attack models can be integrated into real-world applications. By stacking diverse baseline models, we aim to create a multifaceted attack strategy that capitalizes on the varying poison generation techniques employed by each model.

Moreover, we introduce a multi-armed bandit approach on top of the attack cluster to intelligently and effectively remix the poisoned gradients. This bandit approach presents several advantages. First, it enables adaptive selection of the most effective poison generation, considering the state of defense mechanism at the FL server. Second, by amalgamating various attack methods through the bandit strategy, our approach attains a higher level of poison potency, thereby creating a more formidable and adaptable attack. Lastly, and significantly, the integration of the bandit-based technique empowers our proposed attack method to circumvent detection by the defense protocol implemented within the FL server. This amalgamation of baseline attack models and the bandit-based strategy represents an innovative approach that aims to address the limitations of conventional attacks in FedRS [10], [11], [12]. It leverages the adaptability and diversity of multi-armed bandits to create an undetectable yet potent poisoning strategy.

Our contribution centers on introducing an impactful poisoning attack that collectively addresses three fundamental limitations prevalent in existing attack methodologies. Specifically, our approach successfully circumvents the defense mechanisms embedded within the FL server, operates effectively even with a limited number of malicious clients, and enables the targeted promotion of specific items with prior knowledge of FedRS. Through extensive experiments on three benchmarks, our attack strategy has demonstrated noteworthy effectiveness. Our

theoretical analysis and empirical findings underscore a critical inadequacy in the current defensive measures, revealing their insufficiency in detecting vulnerabilities introduced in our attack methodology. This observation serves as a clear indication of the pressing need for enhanced security enhancements to counter our specific attack method within FedRS. In summary, our contribution can be outlined as follows:

- To our best knowledge, we are the first to introduce a poisoning attack designed explicitly for federated recommender systems that remains effective even against recent defense mechanisms, operates proficiently with a limited number of malicious users, and notably, functions without the necessity of background knowledge. This underscores the risks associated with fostering popularity bias and potential backdoor activities within federated environments.
- Our attack strategy uncovers the security risk of recommenders even under privacy preserved FL environment with a small fraction of malicious clients. It raises several question marks on the performance of FedRS and therefore, demands attention on enhancing the security measures.
- We perform extensive experiments on three benchmark datasets, compare attack effectiveness with state-of-the-art model poisoning approaches, and analyze the impact under recent defense mechanisms.

The subsequent sections are structured as follows; Section II presents related work. Section III formally articulates the problem statement, elucidating fundamental definitions and delineating the specific attack settings considered within this study. Section IV meticulously delineates our devised attack approach, detailing components such as the base recommender, the assembled attack cluster, and the integration of a bandit model. Section V provides a detailed exposition of experimental findings. Finally, Section VI presents conclusive remarks and offers insights into potential future research directions.

II. RELATED WORK

In order to market goods or services to users of interest, recommendation models have been developed and widely used. Meanwhile, it has been tried to poison models to skew the results and obtain a covert financial advantage. Therefore, the security and privacy issues have been broadly addressed in the RS literature [18]. Various frameworks, including differential privacy, secure multi-party computation, homomorphic encryption, and federated learning have emerged as potential solutions to address the privacy and security issues prevalent in RS [19], [20]. Despite advancements, the growing use of RS has spurred diverse attack strategies, challenging their fair use. This section analyzes these attack methods, distinguishing between general federated attacks and those specific to FedRS.

A. Attack Against Recommender Systems

Recommender systems are vulnerable to various attacks that bias their outputs, targeting user-item interactions, profiles, and algorithms [4], [8]. Malicious actors often use fake users to promote or demote items, with techniques such as shilling attacks proving effective against collaborative filtering [5], [13].

Beyond data injection, model poisoning attacks also pose significant threats, particularly in federated settings, by manipulating model training to achieve targeted outcomes [13]. Label flipping is a widely used method that does not require background knowledge of training data [9]. Another common data poisoning strategy is generating adversarial samples [21]. However, these attacks are typically conducted on centralized recommendation systems with known training datasets [18] and are less effective in decentralized (federated) settings.

B. Attacks Against Federated Recommenders

The decentralized nature of FedRS introduces unique vulnerabilities exploitable by adversaries. Recent literature [22], [23], [24] has extensively investigated diverse attack strategies, broadly categorized as follows.

1) *Data Poisoning Attacks*: In recent years, various studies have delved into the issue of poisoning attacks within the federated learning paradigm [1], [10]. Data poisoning attacks aim to manipulate the training data in FedRS models to compromise performance or steer recommendations toward malicious goals. These attacks involve injecting malevolent data, such as fake users or manipulated sources, into the training set [25], [26], modifying existing data points, or influencing user behavior to generate biased data. Kasyap and Tripathy [25] investigated vulnerabilities associated with data poisoning attacks in general FL models, while Sun et al. [27] and Zhang et al. [28] highlighted the influence of label manipulation, profile injections, and data tampering techniques on the recommendation process in data poisoning scenarios.

2) *Model Poisoning Attacks*: Targeting the FedRS model itself, model poisoning attacks aim to inject backdoors, alter model parameters, or manipulate the aggregation process. These attacks exploit vulnerabilities in the model training algorithm or communication protocols to compromise the model's integrity. Rong et al. [11], [12] proposed an effective method to poison FedRS with limited data samples. To counter such attacks, Barroso et al. [29] proposed an attacks-resilient federated aggregation method based on robust filtering of outliers. Furthermore, studies by Yang et al. [30] and Shi et al. [31] identified parameter manipulation as a pivotal strategy behind model poisoning attacks.

3) *Evasion Attacks*: Evasion attacks aim to exploit limitations in the FedRS model or manipulate decision boundaries to evade detection or influence recommendations. These include tactics such as strategically hiding items or user behavior from the model, manipulating ratings, or crafting adversarial examples. For instance, attackers employ item hiding strategies to avoid interactions with detrimental items [32], [33] and manipulate ratings to influence item rankings [3], [19]. Crafting adversarial examples involves manipulating user profiles or item data to skew the model's predictions [16], [19].

4) *Hybrid Attacks and Defenses*: On the defensive side, anomaly detection [4], robust learning algorithms [19], and secure aggregation protocols [29] have been explored. However, existing defenses display limited success against adaptive attacks [25]. Continual research is critical to address emerging

threats such as backdoor and inference attacks as well as stronger poisoning attacks that evolve to bypass defenses.

According to existing literature, FedRS attack models are typically categorized into targeted, untargeted, and backdoor attacks [3], [7]. Our work focuses on targeted attacks, specifically designed to promote specific items within the recommendation system. Unlike the existing methods which either suffer from significant accuracy degradation [4], [15] or require a high proportion of malicious users [10], our proposed attack strategy is effective in several aspects. It does not rely on full knowledge of the target recommender, making it more feasible in real-world scenarios where data access is limited. Additionally, our method maintains high attack effectiveness and stealthiness without compromising recommendation performance, even with a minimal number of malicious clients.

III. PROBLEM FORMULATION

In this section, we first revisit the fundamental settings of FedRS and then define our research problem. Also, we formally define the base recommender and basic settings of our attack model.

A. Learning in Federated Environment

Generally, given a set of N users $U = \{1, 2, 3, \dots, N\}$ and a set of M items $I = \{1, 2, 3, \dots, M\}$, the users' feedback for items can be represented by an $N \times M$ preference matrix R where r_{ui} is a rating value stated by user u for item i . For the standard matrix factorization model [34], any rating preference can be estimated as:

$$r_{ui} \sim p_u^\top q_i. \quad (1)$$

In general, a collaborative filtering model obtains r_{ui} through a linear combination of low-dimensional latent user-factors p_u for $u = 1, 2, 3, \dots, N$ users and item-factors q_i for $i = 1, 2, 3, \dots, M$ items collected in factor matrices P and Q respectively. In federated settings, the model performs local training on edge device and shares only model updates with federated server. The goal of a federated recommender is to learn a scoring function $\Omega(u, i)$ to recommend a list of k items to a user u_i . Thus,

$$r_{ui} = \Omega(u, i) = P \cdot Q, \quad (2)$$

where r_{ui} is the predicted score, p_u and q_i are the user and item representation vectors for the user u and item i respectively. Both the user and item vectors are derived from the user encoder $f(u|\Theta_U)$ and the item encoder $f(i|\Theta_I)$ with Θ_U and Θ_I being the user and item parameters. Also, the user or item encoders are agnostic to model, which can be the embedding matrix [35], graph neural network [36], or other models.

Why are the embedding vectors significant for FedRS? Embedding vectors are significant for federated recommenders because they provide a compact and meaningful representation of users and items for the core recommendation algorithm. In FL, these are crucial because they enable the collaboration of multiple users' devices in training the recommendation model. Also, the recommender system gives a user u the personalized

item ranking $\mathcal{R}_u$ during testing:

$$\begin{aligned}\mathcal{R}_u &= \{(i, \text{rank}(r_{ui}))\}_{i \in I} \\ &= \{(i, \text{rank}(p_u^\top \cdot q_i))\}_{i \in I} \\ &= \{(i, \text{rank}(\|q_i\| \cos(p_u, q_i)))\}_{i \in I},\end{aligned}\quad (3)$$

where rank is a ranking function based on a set of scoring values between $u \in U$ and $i \in I$. The personalized ranking is affected by only two factors - the magnitude of the item embedding q_i and its cosine similarity to the user embedding p_u . This is because the ranking function is based on a range of values for scoring functions. Therefore, $\|p_u\|$ is excluded from (3). In particular, the local model gradients for updating a matrix factorization model for observed r_{ui} and unobserved r_{uj} ratings are:

$$\frac{\partial (r_{ui} - r_{uj})}{\partial \Theta} = \begin{cases} (q_i - q_j) & \text{if } \Theta = p_u \\ p_u & \text{if } \Theta = q_i \\ -p_u & \text{if } \Theta = q_j \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

In the federated setting, a popular item i that aligns with mainstream preferences is more likely to be updated by a positive gradient p_u during the local training process. When the item i is used as a negative sample, it will be updated by a negative gradient $-p_u$. During the aggregation of gradients on the central server, the positive and negative gradients from all user devices are combined to generate an updated global model. The frequent positive updates to a particular item i are highly vulnerable in FedRS. Therefore, the magnitude and direction of user-item embedding are highly significant from the attacker's perspective.

B. Base Recommender

Considering the practical significance, we adopt Neural Collaborative Filtering (NCF) [37], [38] as the base recommender. NCF has been widely adopted recommender in e-commerce, streaming services, and social media platforms due to its effectiveness in handling large-scale datasets and enhancing recommendation precision, as evidenced by recent works in the field [1], [2], [5], [39]. In particular, Rendle et al. [40] highlight the comparative advantages of NCF for many practical applications. For formal understanding, let each user $u \in U$ have an embedding vector p_u which represents its latent features. Similarly, each item $i \in I$ has an embedding vector q_i . We use r_{ui} as predicted score between user u and item i , which can be formulated as:

$$r_{ui} = \Omega(p_u, q_i), \quad (5)$$

where Ω is a scoring function used to predict r_{ui} . To model feedback signals with a high-level of non-linearities, NCF utilizes a Multi-Layer Perceptron (MLP) to learn the function Ω . The predication function Ω can be defined as follows:

$$\Omega(p_u, q_i) = a_{out}(\mathbf{h}^T \Psi(p_u \odot q_i)) \quad (6)$$

where a_{out} , $\mathbf{h}$, Ψ , and $\odot$ denote the activation functions, the edge weights of the output layer, the MLP function, and element-wise dot product of vectors respectively.

C. NCF in Federated Environment

In FL settings, the NCF model can be used by allowing each user to train a lightweight model locally. These local models are then sent to a central server, where they are aggregated to generate a global model. We utilize the aggregation step as advised in standard Federated Neural Collaborative Filtering (FNCF) [38]. Details related to the FedAvg algorithm in FedRS, including local and global model parameter settings, can be explicitly found in [39].

Formally, let P and Q denote the embedding matrices of users and items respectively, where $p_u \in P$ and $q_i \in Q$ represent a particular user or item in matrices P and Q . Let $\mathcal{D}_u$ denote the training dataset of a particular user u , which consists of item-score (i, r_{ui}) . If user u has interacted with item i , $r_{ui} = 1$, and $r_{ui} = 0$ otherwise. It is important to consider that for each user u , a large number of items remain uninteracted in I . Thus, the negative instances in $\mathcal{D}_u$ are sampled at a ratio of $b : 1$, meaning that for each positive instance, there are b negative instances. In the federated settings for the NCF model, each user u stores p_u , local model updates $\nabla \Theta$, and $\mathcal{D}_u$ locally, while global model parameters Θ are stored on the FL server. To train the recommender model, we use the Binary Cross-Entropy (BCE) loss to measure the difference between the predicted scores of the model and the ground-truth scores on the training dataset. Thus, (7) formally shows the loss function $\mathcal{L}_u$ for predicted ratings r_{ui} and actual ratings scores r_{ui} :

$$\mathcal{L}_u = - \sum_{i, r_{ui} \in \mathcal{D}_u} (r_{ui} \log r_{ui} + (1 - r_{ui}) \log (1 - r_{ui})). \quad (7)$$

For each federated iteration t , a subset of users $U_t \in U$ is randomly selected by the central server to participate in updating global model parameters Θ_t . After computing $\mathcal{L}_u$, each user locally updates the user factors p_u using a learning rate η . Gradients (∇p_u^t and $\nabla \Theta_u^t$) are derived to represent the direction and magnitude of the change needed to improve the model's performance. The federated learning approach protects users' private data by keeping it on their devices. However, the significance of gradient updates infers that they are potentially vulnerable to attacks. The proposed attack strategy generates poisoned gradients that can effectively compromise the target model and promote the selected item. In our federated settings, the p_u update for the next federated iteration $t + 1$ can be defined as follows:

$$p_u^{t+1} = p_u^t - \eta \nabla p_u^t. \quad (8)$$

The selected user uploads $\nabla \Theta_u^t$ to the federated server for global model update. Finally, after collecting all the uploaded gradients from the selected users, the federated server updates Θ by aggregating the uploaded gradients using the process:

$$\Theta^{t+1} = \Theta^t - \eta \sum_{u \in U_t} \nabla \Theta_u^t. \quad (9)$$

IV. THE PROPOSED APPROACH

In this section, we introduce the details of our HidAttack approach with a multi-armed bandit model. A general overview

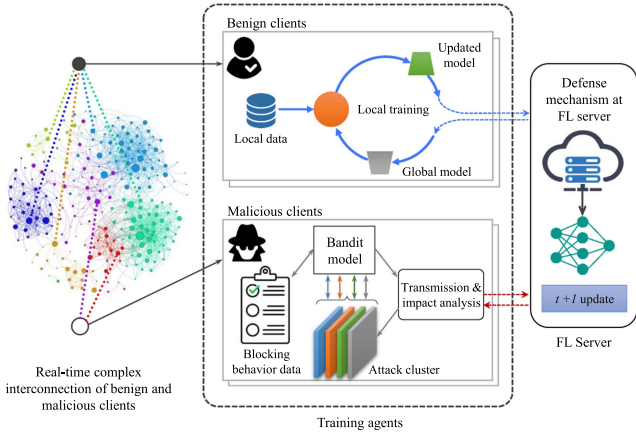


Fig. 1. Overview of the HidAttack strategy: illustrating a complex network, interconnecting benign and malicious clients. Benign clients engage in the FL process, whereas malicious clients leverage a bandit model, employing a cluster of attacks to poison the global model to promote targeted items. The bandit model's role is crucial, generating a variety of undetectable poisoned gradients, evading detection by the aggregation server.

of HidAttack is illustrated in Fig. 1. First, we outline the poisoning attack and HidAttack as an optimization problem, then we discuss the basic attack settings, the attacker's knowledge, and the attacker's capacity. Finally, we describe the attack cluster and bandit details.

A. Poisoning Attacks

In FedRS, poisoning attacks can be broadly classified as untargeted or targeted. These attacks are carefully planned and executed by malicious users participating in the training process. The attack is usually scheduled to start at a particular epoch t , and prior to that, all malicious users participate normally in the training process. Once the attack begins, the malicious users start injecting poisoned gradients by replacing the real local gradients $\nabla\Theta$ with purposefully crafted gradients $\bar{\nabla}\Theta$. This gradual modification of the global model leads to an increase in the popularity of the attackers' targeted items. Detecting targeted poisoning attacks is much more challenging than untargeted attacks. This is because the poisoned updates are carefully crafted to appear as normal updates, and their impact is not easily visible during the training phase. Generally, the attack task can be defined as follows:

Definition 1 (The attack task): Given a FedRS parameterized by Θ , our adversary aims to promote a target item $\hat{i} \in I$ by altering the local gradients submitted by malicious user $\hat{u} \in U$ at each federated iteration t , i.e., $f: \nabla\Theta \rightarrow \bar{\nabla}\Theta$. Note that each malicious user will produce its unique deceiving gradients $\bar{\nabla}\Theta$. Thus, the malicious users aim to learn the optimal deceptive gradients $\bar{\nabla}\Theta_{\hat{u}}$ that will influence the global model to recommend the target item $\hat{i}$ with a higher probability for each benign user. These deceptive gradients are learned by optimizing the objective function, which might be a differentiable function of the model parameters. Once the optimal deceptive gradients are learned, the malicious users can use them to replace the real gradients during the training process.

In similar settings, attacks are presented in academic literature. However, a key issue is neglected in most existing works that an adversary in real-world systems cannot control a significant proportion of users such as 10% to 40% [10]. Even some attack strategies [12], [18], [36] have been designed for a small proportion of users, such as 1% to 10% are still impractical because malicious users can be detected by the defense protocols running at the FL server.

B. Basic Attack Settings

Attack Goal: The attacker aims to bias the global model in the FL environment for promoting a targeted item (targeted attack). The effectiveness of the attack model in FedRS is usually evaluated using the exposure ratio at $K(ER@K)$ metric. The attacker's goal is to increase the $ER@K$ result by making target items $\hat{i}$ appear in the top- K recommendation list. The notation K denotes the number of recommended items and $ER@K$ is the fraction of users whose recommended items include the target item $\hat{i}$. Let the system recommend a list of items $\bar{I}_u \in I$ to a user u from users' unseen items $I_u \in I$ such that $\bar{I}_u \subset I_u$. The selection of $\bar{I}_u$ is made on the maximum prediction score of $I_u \in I$. The exposure ratio at K of the targeted items can be defined as:

$$ER@K = \frac{1}{N} \sum_{u \in U} \frac{|\hat{i} \cdot \bar{I}_u|}{|\hat{i} \cdot I_u|}. \quad (10)$$

Attacker's knowledge: Since the standard aggregation operations are known in FedRS, we assume the attacker has partial knowledge of the aggregation process at the FL server. This is a reasonable assumption because the attackers participate in the FL process as benign user. It enables them to observe the model structure, learning algorithm, and global model parameters updated in the iterative communication rounds, thus constructing the deceptive gradients. However, this assumption applies only to a proportion of malicious clients U_m , not to all participants in the FL process. Unlike most existing attack models in federated settings [7], [8], our HidAttack does not require the attacker to know the global distribution of the training dataset, deep neural network structure, and loss function, or how other benign users are generating local gradients, etc. Therefore, HidAttack would be more effective and obscure to the defense mechanism.

Attacker's capability: It refers to the extent to which the attacker can influence the FL process. Since the effectiveness of the poisoning is largely determined by the number of malicious users (U_m), we assume that the attacker can inject at least a specific percentage (1%) of malicious users into the training process which is significantly lower than in existing studies [10]. As a malicious user in the training process, the attacker can arbitrarily modify the global model gradients. When the victim model consumes these bogus users, the attacker will achieve the nefarious objective. This shows that the actual poisoning occurs during the victim model's training phase as opposed to its test time.

C. HidAttack as an Optimization Problem

Different from heuristic approaches [18], we design HidAttack as a standard optimization problem. The key objective is to maximize Exposure Ratio (ER) of the target item in top- K recommendations, by manipulating model updates over T iterations. The undetectability constraint prevents attackers from being flagged by the server defense mechanism. The size and diversity constraints maintain a limited, evolving attacker profiles. The anomaly avoidance bound restricts manipulations to avoid detection. The adversaries will alter the global model gradients for maximizing $ER@K$ while consistently deceiving the defense mechanism at the FL server. Suppose U_m and $\nabla\Theta^t$ represent the set of malicious users and the corresponding poisoned gradients at federated iteration t , respectively. Let T denote the total number of training iterations and U^t , Θ^t , and Θ^t denote the parameters U , S , and Θ at the end of t -th iteration, respectively. Then we can formally define the objective as:

$$\begin{aligned} & \max_{(\nabla\Theta^{t \in T})} ER@K(U^t, \Theta^t, \hat{i}), \\ & s.t. \quad \begin{cases} \hat{u} \neq U_{\hat{u}}, & (\forall \hat{u} \in U_m) \\ |\hat{u}| \leq |U_m|, \\ \nabla\Theta^{t+1} \neq \nabla\Theta^t & (\forall t \in T). \end{cases} \end{aligned} \quad (11)$$

where U_m denotes the set of malicious users injected by attacker and $U_{\hat{u}}$ represents the set of users recognized as malicious by the FL server. The notation $|U_m|$ denotes the upper bound of the proportion of malicious users allowed to participate in training process.

D. Attack Cluster

In this section, we present detailed description of our HidAttack, a method for poisoning a federated recommender by factiously promoting targeted items. Our model poisoning framework to federated neural collaborative filtering model involves stacking four well-known poison models $\mathcal{A}_1$ [16], $\mathcal{A}_2$ [10], $\mathcal{A}_3$ [12], and $\mathcal{A}_4$ [17] as baseline attack models for item promotion in federated settings. Primarily, these methods are designed to generate poisoned gradients that, when aggregated with global model by a malicious user during federated learning, the model will recommend certain items to users. The method $\mathcal{A}_1$ [16] proposed a triple adversarial learning approach for influence-based poisoning attacks. It generates malicious users through triple adversarial learning of the generator, discriminator, and influence module, and explores an effective approximation approach for estimating each fake user's influence. The second method $\mathcal{A}_2$ Explicit Promotion (EP), is a part of the PipAttack model [10] that backdoors a similar FedRS for manipulating targeted item promotions. It takes advantage of the inherent popularity bias in data-driven recommenders and enables the target item to have the characteristics of popular items in the embedding space. The third method $\mathcal{A}_3$ [12], proposed a random approximation strategy to generate poisoned gradients FL settings. This strategy involves randomly selecting benign users' embedding vectors and perturbing them to create poisoned gradients. The fourth

method $\mathcal{A}_4$ [17] described a way for poisoning a factorization-based collaborative filtering model by injecting fake user-item interaction data.

A malicious user $u \in U_m$ utilizes the attack cluster $\mathcal{C}_A$ to generate poisoned gradients. Each attack within this cluster is implemented according to the standard settings described in the original publications, as cited in the previous paragraph. Based on the bandit reward $\mathcal{R}_t$ (formally described in Section IV-E), each malicious user adaptively selects the most effective poisoned gradients to maximize the exposure ratio in the global model. This selection mechanism ensures that gradients with the highest reward contributions are prioritized, allowing targeted item exposure to increase iteratively with minimal detection risk. By combining these attack strategies, the bandit model intelligently selects and remixes poisoned gradients from the four methods to bypass the defense mechanism at the FL server. This ensemble approach enhances attack effectiveness by leveraging the unique characteristics and diverse gradient generation methods of each attack model, making the combined effect more impactful than any single model. The bandit model thus refines poisoned gradients over time to maximize the exposure ratio.

E. Bandit Model

On top of the attack cluster, we design a bandit approach that intelligently selects the effective poisoned gradients. Given the four baseline attack models, i.e., $\mathcal{A}_1$, $\mathcal{A}_2$, $\mathcal{A}_3$, and $\mathcal{A}_4$, a bandit algorithm [41] is used to remix and choose the most effective combination of gradients from the baseline poison. The bandit model intelligently selects a subset of poisoned gradients such that the defense mechanism at FL server could not detect the poisoned pattern. Formally, given a set of users $U_t = U_b \cap U_m$ (contain both benign and malicious users) participating in a particular federated round t , each client has equal opportunity to update the global model parameters $\nabla\Theta$. Each malicious client $\hat{u}$ contains a cluster of attack models $\mathcal{C}_A = \mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_k$ that can generate poison gradients subject to promote a targeted item $\hat{i}$. Initially, each malicious client $\hat{u} \in U_m$ runs the baseline attacks and independently submits poisoned gradients to the Multi-Arm Bandit (MAB) model for effective poison recreation. The bandit model evaluates each poisoned gradient by analyzing the potential exposure impact of each attack model under federated training dynamics, allowing a flexible yet covert attack pattern across rounds. The gradient aggregation process for poison generation at federated iteration t can be formally expressed in (12):

$$\mathcal{P}_t = \sum_{u \in U_t} \sum_{a \in \mathcal{C}_A} w_a \mathcal{P}_{u,a}, \quad (12)$$

where w_a represents the weight associated with each baseline attack in the cluster $\mathcal{C}_A$. Additionally, $\mathcal{P}_{u,a}$ denotes the poisoned gradient generated by a specific malicious client $u \in U_m$ using attack model $a \in \mathcal{C}_A$. The selection of an effective poison arm depends on the reward $\mathcal{R}_t$, which is calculated based on the impact of each poisoned gradient on the global model's loss. This reward guides the bandit model to prioritize attack models that yield the most effective poison generation in each federated

round. Each poisoned gradient $\mathcal{P}_{u,a}$ is dynamic and crafted to subtly influence the global model's parameters to promote the targeted item while remain undetectable by the defense mechanism. Formally, $\mathcal{P}_{u,a}$ can be expressed as:

$$\mathcal{P}_{u,a} = \nabla\Theta_u + \overline{\nabla}\Theta_a, \quad (13)$$

where $\nabla\Theta_u$ is the benign gradient that a client u would typically submit, and $\overline{\nabla}\Theta_a$ is the perturbation introduced by the attack model $a \in \mathcal{C}_A$, designed to manipulate the global model's behavior. Each $\overline{\nabla}\Theta_a$ ($\forall a \in \mathcal{C}_A$) is crafted to maximize the exposure of the target item while remaining within the bounds of plausible client updates. During each federated round, the bandit model generates a varied composition of poisoned gradients by adjusting the weight vector w dynamically, thereby concealing the pattern of malicious updates and making detection more challenging. Specifically, in each federated iteration $t \in T$, the selection of poisoned gradients is informed by a reward $\mathcal{R}_t$ that quantifies the impact of the previous gradient on the global model's effectiveness in promoting the targeted item. The reward $\mathcal{R}_t$ is computed by examining the difference in model loss before and after applying a poisoned gradient, with Θ_{t-1} and Θ_t representing the global model parameters at iterations $t-1$ and t , respectively. To formally capture this impact, we define the instantaneous loss difference as $\mathcal{R}_t$ at iteration t as:

$$\mathcal{R}_t = \ell(\Theta_{t-1}, \mathcal{L}) - \ell(\Theta_t, \mathcal{L}), \quad (14)$$

where $\mathcal{L}$ denotes the loss function governing the base recommender, and ℓ represents the observed loss at reward function. This loss difference serves to calculate the reward $\mathcal{R}_t$ in the bandit model, which encourages the selection of poison gradients that lead to the greatest reduction in model loss while continuously promoting the targeted item. Additionally, this reward structure ensures that poisoned gradients achieving higher reductions in model loss are favored in subsequent iterations, promoting an effective yet covert poisoning strategy. As outlined in (12), the weight vector w plays a vital role in guiding the selection of poisoned gradients that are most effective. By adjusting the weights based on the calculated reward $\mathcal{R}_t$, the vector w emphasizes gradients that yield higher reductions in model loss, thereby enhancing the exposure of the targeted item. This adaptive adjustment, achieved through an exponential decay function, assigns greater importance to gradients with higher exposure ratio (under reward $\mathcal{R}_t$) while dynamically balancing exploration and exploitation within the bandit model. The weight update is formally defined as:

$$w_a(t+1) = w_a(t) \exp\left(\frac{\bar{\eta}\mathcal{R}_t\mathcal{P}_{t,a}}{\sum_{j \in \mathcal{C}_A} w_a(t)\mathcal{P}_{t,j}^2}\right), \quad (15)$$

where $\bar{\eta}$ is the learning rate to observe the diverse impact of poison on associated reward $\mathcal{R}$ at t th iteration. The intuition behind this rule is passing higher priority to the poison gradients $\mathcal{P}$ that result in a larger decrease in the global model loss, while $a \in \mathcal{C}_A$ that generate less effective poison gradients will receive lower weights. The exponential weight algorithm ensures that the weights are updated in a way that balances exploration (i.e., trying out different attack models $a \in \mathcal{C}_A$) and exploitation (i.e.,

focusing on poison gradients with higher rewards). Therefore, the bandit model on top of the attack cluster allows the dynamic poison generation from the baseline attack model to generate the most effective and undetectable poison gradients.

F. Discussion

We provide rigorous analytical arguments regarding the key properties that enable HidAttack efficacy and evasion capabilities within FedRS. Before presenting formal arguments, it is crucial to understand that HidAttack effectively integrates both model poisoning ($\mathcal{A}_2$ and $\mathcal{A}_3$) and data poisoning ($\mathcal{A}_1$ and $\mathcal{A}_4$) in federated settings. While $\mathcal{A}_2$ and $\mathcal{A}_3$ modify gradients directly in a decentralized environment, $\mathcal{A}_1$ and $\mathcal{A}_4$ originally designed for centralized systems are adapted to the federated context by generating synthetic user profiles locally on each malicious client. This local manipulation impacts item popularity, aiding in the generation of poisoned gradients that evade server-based defenses. These gradients are then strategically combined using our bandit model, allowing HidAttack to exploit the strengths of both model poisoning and data poisoning approaches.

Additionally, the effectiveness of HidAttack stems from its novel use of a multi-armed bandit model. First, the bandit model enhances stealthiness by dynamically adapting the base attack strategies, making malicious patterns less predictable and harder to detect. Second, it optimizes attack effectiveness by learning to prioritize the most successful methods in various scenarios, allowing the system to maintain potency even if individual attack vectors become less effective over time. While the ensemble approach contributes to the attack's strength, it is the intelligent combination and optimization of these diverse methods that truly sets HidAttack undetectable. The bandit model's learning mechanism exploits the strengths of each attack while mitigating their individual weaknesses, creating a synergistic effect that surpasses the mere sum of its parts. This adaptive approach not only improves the attack's success rate but also significantly increases its resilience against evolving defense mechanisms, making HidAttack effective across a range of scenarios and recommender system configurations.

To formally demonstrate the effectiveness within the ϵ -greedy MAB architecture, consider a FedRS consisting of $U_t \subset U$ clients participating in each federated round t . Each client u_i holds a local model Θ_i representing its preferences. HidAttack employs an MAB model to select from k diverse baseline attack models $\mathcal{A}_k$, each generating poisoned gradients $g_k(\Theta_t)$. The attacker aims to maximize the Exposure Ratio (ER) of a target item i in the recommendations generated by Θ_t . Formally, following theorems and lemmas witness the convergence of bandit selection and undetectability of poisoned gradients in the proposed attack strategy.

Theorem 1 (Effectiveness of Poisoned Gradients): Consider each baseline attack model $\mathcal{A}_k$ generates gradients $g_k(\Theta_t)$ that monotonically increase ER_i within a bounded range. For any arm $\mathcal{A}_k$ with $E_k(\Theta_t) > E_{A_s}(\Theta_t)$, applying $g_k(\Theta_t)$ at round t will result in $E_k(t+1) > E_{A_s}(t+1)$.

Proof: The gradient $g_k(\Theta_t)$ guarantees a positive increase in ER_i compared with the global model without the poisoned

gradient. With the learning rate $\bar{\eta}$, the updated estimate $E_k(t+1)$ will reflect this improvement:

$$\begin{aligned} E_k(t+1) &= (1 - \bar{\eta}) E_k(t) + \bar{\eta} (ER_i(\Theta_t + g_k(\Theta_t))) \\ &> E_k(t) + \bar{\eta} \Delta_{k,t}, \end{aligned} \quad (16)$$

where $\Delta_{k,t}$ is the positive increase in ER_i due to $g_k(\Theta_t)$. Similarly, $E_{A_*}(t+1)$ will also increase:

$$\begin{aligned} E_{A_*}(t+1) &= (1 - \bar{\eta}) E_{A_*}(t) + \bar{\eta} (ER_i(\Theta_t + g_{A_*}(\Theta_t))) \\ &> E_{A_*}(t) + \bar{\eta} \Delta_{A_*,t}. \end{aligned} \quad (17)$$

However, since $E_k(\Theta_t) > E_{A_*}(\Theta_t)$, we have $\Delta_{k,t} \geq \Delta_{A_*,t}$, leading to the final inequality:

$$\begin{aligned} E_k(t+1) &> E_k(t) + \bar{\eta} \Delta_{k,t} > E_{A_*}(t) + \bar{\eta} \Delta_{A_*,t} \\ &> E_{A_*}(t+1). \end{aligned} \quad (18)$$

This confirms that $E_k(t+1)$ surpasses $E_{A_*}(t+1)$, effectively demonstrating the advantage of exploiting arms with higher expected ER_i increases. $\square$

Theorem 1 formally shows that the poisoned gradients from each baseline attack model provably increase exposure ratio for the target item. However, this alone does not ensure the overall attack strategy will select the optimal model. To address this limitation, we state another argument that formally proves whether the bandit-based selection provably converges to the highest utility arm. While the initial argument (Theorem 1) clarifies the effectiveness of poisoned gradients to monotonically increase the target item's ER_i , it does not guarantee optimal selection. Theorem 2 directly addresses this limitation by formally proving the bandit model's convergence to the best arm. In essence, it fills the logical gap and highlights how the attack judiciously identifies the most effective model for crafting poisoned gradients. Together, Theorem 1 and Theorem 2 establish a complete two-step argument - poisoned gradients are potent, and bandit selection reliable, thereby formally validating HidAttack's robust improvement of exposure ratio.

Theorem 2 (Convergence of Bandit Selection): Given, the expected ER_i for each arm ($E_k(\Theta_t)$) possesses finite variance. With probability $1 - \delta$, where δ serves as an upper bound on the confidence level, the ϵ -greedy MAB policy converges to selecting the arm with the highest expected ER_i (denoted as A_*) within a finite number of rounds.

Proof: The ϵ -greedy algorithm ensures exploration with probability ϵ and exploitation with probability $1 - \epsilon$. Considering the finite variance assumption, the estimates $E_k(t)$ for each arm converge to their true expected values with high probability. Consequently, after a sufficient number of rounds, the MAB policy consistently opts for the arm with the highest expected ER_i , i.e., A_* . This assertion can be formalized utilizing Hoeffding's inequality with appropriate confidence bounds [42]. $\square$

Theorem 3 (Maximizing ER_i with High Probability): There exists at least one arm A_k with $E_k(\Theta_t) > E_{A_*}(\Theta_t)$ for any initial state Θ_t . Thus, the ϵ -greedy MAB policy implemented in HidAttack guarantees, with probability at least $1 - \delta$, that ER_i will reach a value closer to its maximum achievable value than the initial ER_i within a finite number of rounds.

Proof: Combining Theorem 1 and Theorem 2, the MAB policy converges to selecting arms with higher expected ER_i . Given the existence of at least one arm (A_k) initially offering a better increase than the optimal arm (A_*), the policy is likely to prioritize A_k . As these gradients are applied, ER_i will consistently trend toward its maximum attainable value within the system's constraints. By Hoeffding's inequality [42], the deviation from the expected maximum ER_i is bounded with confidence $1 - \delta$, thus demonstrating the effectiveness of the proposed attack in maximizing exposure ratio. $\square$

In addition, the undetectability of poisoned gradients in HidAttack is significant. For formal insights, we analyze its ability to evade common federated server anomaly detection mechanisms. Here, we present two key arguments in Theorem 4 and Lemma 1.

Theorem 4 (Obfuscation through Bandit-based Diversification): The baseline attack models in HidAttack generate diverse poisoned gradients with distinct patterns and effects. With sufficiently diverse baseline models and effective MAB selection, HidAttack can obfuscate the server's anomaly detection algorithms.

Proof: Static anomaly detection in FL applications typically relies on identifying deviations from expected model updates or client behavior [4]. HidAttack's MAB framework continuously explores and utilizes diverse attack models, generating gradients with varying patterns and impacts on the global model. Theorem 1 provides a formal proof that this diversification makes it challenging for static anomaly detection algorithms to establish a stable baseline for identifying malicious behavior. Additionally, the MAB's adaptive selection further obfuscates the origin of poisoned gradients by constantly shifting between different attack strategies. As a result, even if the server detects irregular updates within individual rounds, subsequent alterations in poison generation often obscure the defense mechanism. Additionally, bandit based diversified and evolving gradients play a vital role in obfuscation. Lemma 1 further supports the claim through a different perspective. $\square$

Lemma 1 (Evolving Gradients Mitigate Detection Mechanism): *Assumption:* The poisoned gradients generated by HidAttack are designed to adapt and evolve over time, gradually influencing the global model. *Statement:* Evolving poisoned gradients significantly reduce the effectiveness of static anomaly detection mechanisms based on fixed thresholds or anomaly patterns.

Proof: Static anomaly detection often relies on pre-defined thresholds or pre-identified patterns to flag suspicious updates in the FL environment. However, HidAttack's evolving gradients continuously change their characteristics, making it difficult to establish static thresholds or patterns for reliably identifying them. For example, the attack might gradually increase the exposure of the target item over multiple rounds instead of causing a sudden spike, thereby evading detection based on abrupt changes. Furthermore, the inherent noise and variations in client updates within federated architecture provide additional camouflage for the subtle modifications introduced by HidAttack's evolving gradients. $\square$

TABLE I
EFFECTIVENESS OF HIDATTACK WITH VARYING PROPORTIONS OF MALICIOUS USER (U_m)

Dataset	Measure	$U_m = 1\%$	$U_m = 2\%$	$U_m = 3\%$	$U_m = 4\%$	$U_m = 5\%$
Amazon	ER@5	0.80568	0.83693	0.85185	0.86667	0.90000
	ER@10	0.83328	0.86667	0.86631	0.86667	0.90000
	NDCC@10	0.00505	0.00629	0.00602	0.00581	0.00556
	MRR@10	0.00078	0.00086	0.00081	0.00078	0.00079
MI-1M	ER@5	0.50333	0.83339	0.80007	0.83461	0.92329
	ER@10	0.64682	0.83580	0.83585	0.84774	0.92737
	NDCC@10	0.03211	0.02669	0.02669	0.02669	0.02534
	MRR@10	0.00613	0.00646	0.00625	0.00646	0.00646
YM	ER@5	0.00220	0.08238	0.47169	0.71846	0.89983
	ER@10	0.02199	0.46692	0.62142	0.93824	0.95897
	NDCC@10	0.02798	0.02847	0.02814	0.02866	0.02864
	MRR@10	0.01136	0.01153	0.01159	0.01162	0.01163

Results reveal notably improving exposure ratios even with 1-2% malicious users and approaching maximum efficacy at 5%. Both datasets reflect satisfactory recommendation indicators.

The merger of diverse bandit-based attack models and continuously evolving poisoned gradients within our proposed attack strategy suggests a substantial hindrance to the capability of existing defense mechanisms in detecting attack within the federated server. Although these formal arguments provide robust evidence supporting the potential undetectability of our attack model, it is imperative to consider additional factors such as the number of malicious actors involved and the level of control exerted by the attacker over the malicious clients. In this regard, the empirical evidence presented in Table I validates the effectiveness of our proposed attack model, particularly when operating with a limited number of malicious clients. Therefore, while this formal analysis accentuates the potential of HidAttack to circumvent existing defenses, further research and adaptation are indispensable to fortify its resilience against evolving server-side detection mechanisms.

V. EXPERIMENTAL EVALUATION

In this section, we describe the experimental settings including datasets, baseline attack methods, evaluation metrics, and parameter settings. Additionally, we investigate the effectiveness of our attack model under varying conditions and analyze its impact to boost targeted items in presence of state-of-the-art defense methods. Particularly, we aim to answer the following Research Questions (RQs):

- **RQ 1:** Does the proposed HidAttack perform poisoning effectively on selected FedRS?
- **RQ 2:** Is the proposed HidAttack effective under recent defensive protocols for FedRS?
- **RQ 3:** Does the proposed HidAttack significantly affect the recommendation performance of FedRS?

A. Experimental Settings

Datasets: To validate the significance of the proposed HidAttack, three publicly available benchmark datasets are used. For dataset selection, we considered the diversified areas for true validation. We use the Amazon Appliances (Amazon)¹, MovieLens-1M (MI-1 M)², and YahooMusic (YM)³ datasets. In

comparison to Amazon, which has 602,777 ratings from 30,252 individuals and 515,650 Appliances, MI-1 M has 1 million ratings from 6,039 users and 3,705 movies, while YM contains 311,704 ratings from 15,400 users over 1000 musical items. We follow common settings [10], [12] and convert input ratings into implicit feedback. The user feedback has been binarized so that all interacted items had $r_{ij} = 1$ and all uninteracted items got $r_{ij} = 0$ in the rating matrix R . Considering the sparsity of input datasets, the sample size for dividing user-item interactions is taken as $q_i = 4$ negative instances for each positive instance.

Baseline attacks: We compared the attack effectiveness of HiddAttack with the following data and model poisoning strategies in FL environment including:

- **Popular Attack (PA) [7]:** A type of poisoning attack in which Byzantine clients purposefully inject malicious data to raise popularity index of the targeted item. It is a general modal biasing strategy discussed in several existing works [10], [12], [36].
- **Random Attack (RA) [4]:** It works similar to PA, but purposefully injected fake users manipulate both target items and randomly selected items. For large-scale systems, the impact of RA is quite uncertain because targeted promotion mixed with random items selection.
- **Explicit Promotion (EP) [10]:** It is one component of PipAttack that takes advantage of the inherent popularity bias that commonly exists in data-driven recommenders and enables the target item to have the characteristics of popular items in the embedding space. It explicitly enhances the predicted scores of target items for malicious users. More specifically, different from benign users, malicious users take the explicit item promotion objective as their loss function.
- **Hard user Mining (HM) [12]:** HM is a recent effective model poisoning approach. It involves selecting a small subset of users who are difficult to recognize as malicious users. Using characteristics of these users, the attacker generate poisoned gradients. The poisoning strategy works well even if the proportion of malicious users is small.
- **Adversarial Lens (AL) [21]:** AL explores the threat model by a single, non-colluding malicious agent where the adversarial objective is to cause the model to misclassify a set of chosen inputs with high confidence.

¹<https://jmcauley.ucsd.edu/data/amazon/>

²<https://grouplens.org/datasets/movielens/>

³<https://webscope.sandbox.yahoo.com/>

TABLE II
COMPARISON OF HIDATTACK WITH BASELINE ATTACK MODELS UNDER VARYING PROPORTIONS OF MALICIOUS USERS (2%, 4%, AND 5%) ON SELECTED DATASETS

Datasets	U_m	Measure	PA	RA	EP	HM	AL	HidAttack
Amazon	2 %	ER@5	0.00000	0.00000	0.00000	0.00028	0.80089	0.83693
		ER@10	0.00000	0.00000	0.00012	0.33225	0.83333	0.86667
		NDCG@10	0.00624	0.00626	0.00434	0.00613	0.00596	0.00629
		MRR@10	0.00082	0.00081	0.00057	0.00079	0.00078	0.00086
	4 %	ER@5	0.00000	0.00000	0.56667	0.41849	0.83378	0.86667
		ER@10	0.00000	0.00000	0.56667	0.59946	0.85633	0.86667
		NDCG@10	0.00633	0.00625	0.00355	0.00617	0.00587	0.00581
		MRR@10	0.00091	0.00081	0.00075	0.00082	0.00078	0.00078
	5 %	ER@5	0.00000	0.00000	0.66667	0.83693	0.86666	0.90000
		ER@10	0.00000	0.00000	0.66667	0.86667	0.86667	0.90000
		NDCG@10	0.00624	0.00628	0.00358	0.00620	0.00578	0.00556
		MRR@10	0.00081	0.00081	0.00056	0.00086	0.00078	0.00079
MI-1M	2 %	ER@5	0.00000	0.00000	0.00000	0.02164	0.09319	0.83339
		ER@10	0.00000	0.00000	0.00000	0.06571	0.23385	0.83580
		NDCG@10	0.03523	0.03507	0.0353	0.03442	0.03345	0.02669
		MRR@10	0.00646	0.00646	0.00646	0.00646	0.00646	0.00646
	4 %	ER@5	0.00000	0.00000	0.11963	0.04911	0.16753	0.83461
		ER@10	0.00000	0.00000	0.2928	0.13031	0.41293	0.84774
		NDCG@10	0.03523	0.03518	0.03385	0.03400	0.03299	0.02669
		MRR@10	0.00646	0.00646	0.00646	0.00646	0.00646	0.00646
	5 %	ER@5	0.00000	0.00000	0.23927	0.07658	0.24187	0.92329
		ER@10	0.00000	0.00000	0.58560	0.19491	0.59200	0.92737
		NDCG@10	0.03530	0.03122	0.03241	0.03359	0.03254	0.02534
		MRR@10	0.00648	0.00646	0.00646	0.00646	0.00646	0.00646
YM	2%	ER@5	0.00000	0.00000	0.00000	0.05416	0.01255	0.08238
		ER@10	0.00000	0.00000	0.00000	0.44461	0.18923	0.46692
		NDCG@10	0.02817	0.02827	0.02867	0.02727	0.02731	0.02847
		MRR@10	0.01157	0.01158	0.01163	0.01104	0.01105	0.01153
	4%	ER@5	0.00000	0.00000	0.00059	0.74548	0.54777	0.71846
		ER@10	0.00000	0.00000	0.00178	0.93797	0.94752	0.93824
		NDCG@10	0.02817	0.02827	0.02907	0.02675	0.02697	0.02866
		MRR@10	0.01157	0.01158	0.01169	0.01100	0.01104	0.01162
	5%	ER@5	0.00000	0.00000	0.05485	0.88865	0.88133	0.89983
		ER@10	0.00000	0.00000	0.74198	0.94153	0.95338	0.95897
		NDCG@10	0.02831	0.02831	0.02889	0.02673	0.02690	0.02864
		MRR@10	0.01155	0.01155	0.01170	0.01100	0.01104	0.01163

Evaluation protocol: Following the common setting for evaluating attack effectiveness in FedRS [10], [12], we use average Exposure Ratio @ K ($ER@K$), where K is the number of recommended items and $ER@K$ is the fraction of users whose recommended items include the target item. A higher $ER@K$ indicates stronger poisoning effectiveness. Formally, $ER@K$ is defined in (10). To ensure fairness in our experiments, we select the unpopular items (i.e., the items with the least number of interactions) as our target items for both datasets. To observe the recommendation performance, we use $NDCG@10$ and $MRR@10$. To quantify the fraction of observing the ground truth items in the top- K recommendation lists, similar ranking settings are used as discussed in [10], [12].

Implementation details and parameters settings: We implement the proposed method with PyTorch. The architecture of the NCF model is made up of two sub-networks: (1) Multi-Layer Perceptron (MLP) and (2) Generalized Matrix Factorization (GMF). Both the MLP and GMF networks are used to learn the latent factors in the form of user and item embeddings. We employ a 16-hidden layer for MLP with a ReLU activation function. While restricting the size of the latent factors to 4, we set the architecture of the NCF model to be [8, 16, 8], where the first and the last layer is the concatenation MLP and GMF individual layers. The last layer of GMF is obtained by multiplying the user embedding vector with the item embedding vector while the last layer of MLP is obtained by concatenation of the user embedding vector with the item embedding vector. For the user

embedding layer, the input channel is the number of unique users N for each of the experimental datasets and the output channel is the number of factors (set to 4), likewise for the item embedding layer the input channel is number of unique items M of the datasets and output channel is the number of factors (set to 4). The sigmoid activation function is applied after the last layer to transfer the current task as a series of binary classification tasks.

For parameter settings, the model configuration has a batch size 32, and learning rate of $5e-2$. Additionally, we optimize the model with a BCE loss to learn optimal weights, where we take 4 negative instances per positive sample. To ensure consistent settings across all baseline attacks, the same hyper-parameters configurations are employed, which have been identified as optimal in previous studies [10], [12]. Explicitly for HidAttack, we set the learning rate $\bar{\eta}$ to 0.01 for the exponential weight update. The initial weights w_a for each baseline attack model are set to 0.25, ensuring an equal starting probability for all four attacks. We update the weights after each federated round using the exponential decay described in (15). We set the exploration-exploitation trade-off parameter ϵ to 0.075, allowing for occasional random selection of attack models to maintain diversity in the attack strategy. In addition to these details, consider our GitHub repository⁴ for detailed implementation guidelines and dataset preprocessing.

⁴<https://github.com/waqar-uestc/HidAttack>

B. Attack Effectiveness (RQ 1)

In order to evaluate the effectiveness of HidAttack, we conduct experiments with varying proportion of malicious users (U_m) ranging from 1% to 5%. The findings, as summarized in Table I, underscore the robustness of our attack model across different degrees of malicious engagement within the federated learning environment. Notably, even with a mere 1% or 2% representation of malicious users ($U_m=1\%$ or $U_m=2\%$), the attack displayed a substantial increase in $ER@K$, showcasing its notable impact while keeping recommendation indicators (NDCG and MRR) close to the maximum achievable levels. This confirms our initial objective that HidAttack yields considerable effectiveness even with a minimal percentage of adversarial involvement. Furthermore, as the proportion of malicious users escalated, the effectiveness of HidAttack exhibited a pronounced improvement. Particularly, at $U_m=5\%$, the $ER@10$ metric soared to an impressive value of 0.9000, approaching the theoretical maximum. This progression clearly delineates the escalating impact of the attack with an increasing number of malicious actors, showcasing its adaptability and potency in influencing the recommendation outcomes.

Remarkably, our results not only highlight a rapid increase in exposure ratio, but also underscore the satisfactory gain of selected recommendation indicators, i.e., NDCG and MRR, across both the Amazon Appliances and MI-1 M datasets. This multifaceted success serves as a testament to the versatility and effectiveness of HidAttack in not only amplifying exposure but also maintaining the overall recommendation quality. This experimental elucidation aligns with **RQ 1** that HidAttack operates effectively with a minimal percentage of malicious users while displaying an exponential surge in impact with an increase in adversarial engagement. These findings validate the robustness and adaptability of our attack strategy within the federated learning setting.

The comparison presented in Table II underscores the efficacy of HidAttack against various baseline attack models across diverse proportions of malicious users (2%, 4%, and 5%) on both the Amazon and MI-1 M datasets. Notably, the findings reveal HidAttack's consistent advantage over baseline attacks, reflecting its robustness in diverse adversarial settings within FedRS environments. Among the baseline attack models, RA (Random Attack) [4] and PA (Popular Attack) [7] demonstrate limited impact, particularly evident when confronted with a restrained number of Byzantine clients. Similarly, EP (Explicit Promotion) [10] exhibits modest effectiveness, but its influence remains marginal, especially in scenarios with a scarcity of Byzantine actors. The experimentation outcomes unveil a distinct trend: the baseline attack methods tend to amplify the visibility of the targeted item predominantly when a higher proportion of Byzantine clients, such as 5%, is present within the system. Conversely, their effectiveness diminishes significantly when the ratio of malicious clients is restricted (such as 2%), exemplifying their reliance on a higher volume of adversaries to manipulate the recommendation process effectively.

However, amidst this landscape, both HM (Hard user Mining) [12] and AL (Advanced Lens) [21] attacks demonstrate

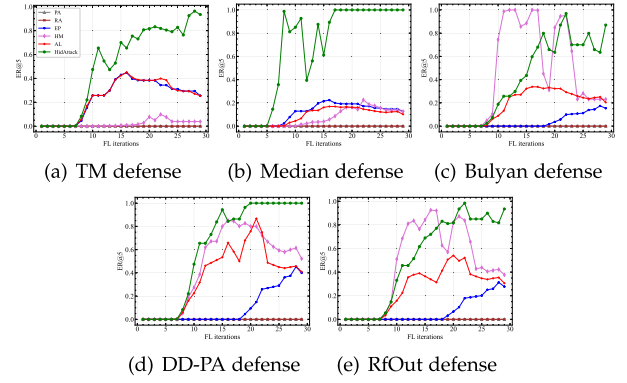


Fig. 2. HidAttack performance under five selected defense protocols. Results were drawn with using 3% malicious users and evaluated attack effectiveness with $ER@5$.

comparatively satisfactory performance even in scenarios with fewer Byzantine clients. This resilience showcased by HM and AL underlines their potential viability in scenarios with limited malicious entities, hinting at a certain adaptability to such constrained adversarial conditions. Crucially, these results underscore the superiority of HidAttack, highlighting consistent and superior efficacy compared with the baseline attack methods across varied malicious user proportions. The comprehensive superiority of HidAttack in influencing recommendations even under conditions of limited adversarial presence reaffirms its potential as an effective strategy for undermining federated learning systems' integrity.

C. Attack Effectiveness Under Defense (RQ 2)

Evaluating how attacks withstand various defense strategies is essential for understanding their true effectiveness and resilience. In this subsection, we make a significant contribution by rigorously analyzing the effectiveness of HidAttack under five distinct defense mechanisms. Our findings, detailed in Fig. 2, demonstrate not only the robustness of HidAttack but also its ability to adapt and maintain high impact across diverse defense paradigms. Each defense mechanism was carefully selected for its unique characteristics, providing a comprehensive assessment of HidAttack's performance in the face of advanced security measures.

- 1) *Trimmed-mean* [43]: This method, centered around retaining updates close to the average for aggregation while excluding extreme values in each dimension, offers a basic yet effective defense approach.
- 2) *Median* [4]: Employing a robust-aggregation strategy by replacing the arithmetic mean with the median of model updates, this defense technique focuses on selecting a value representing the center of the distribution, ensuring resistance against outliers.
- 3) *Bulyan* [44]: Combining MultiKrum [1] and Trimmed-mean, this method sorts clients by geometric distances and utilizes a filter parameter to remove specific clients from the distribution's tails, offering a comprehensive defense mechanism.

- 4) *DD-PA* [44]: An aggregation operator designed to dynamically exclude adversarial clients, preventing the corruption of the global learning model and showcasing adaptability against evolving attacks.
- 5) *RfOut* [29]: This robust aggregation method stands as a resilient defensive mechanism specifically engineered to counter model-poisoning backdoor attacks through the effective identification and filtration of updates from adversarial clients.

Our assessment serves to delve into the effectiveness of proposed attacks in traversing through a diverse landscape of defense strategies. Each defense mechanism presents a unique challenge, from basic outlier handling to dynamic adaptability, shedding light on the agility and adaptiveness of the attacks against an array of defense strategies. The results on the MI-1 M dataset are shown in Fig. 2, and we use 3% malicious users and evaluate attack effectiveness with $ER@5$ to get these observations. The results demonstrate that HidAttack outperforms all baseline attack models under almost all defense methods. In particular, our attack strategy remains highly effective in terms of $ER@5$ under the Median, Bulyan, and DD-PA defense mechanisms. Even under the remaining defense methods, HidAttack outperforms the baseline models. The deviating performance under Trimmed-mean and RfOut might be due to the fact that it filters out a large number of participants in each federated round. In the case of practical RS, such defenses are mostly undesired since they will cause potential loss of recommendation performance.

Overall, our findings demonstrate the superiority of HidAttack over baseline attack models in terms of attack effectiveness and its ability to maintain recommendation performance under different defense strategies. It can be seen that HidAttack cannot be effectively detected by most defense mechanisms. This is because it dynamically injects poisoning gradients and underlines bandit approach continuously observing the impact of poison generation through the exploration-exploitation phenomenon. Our experiments reveal the fact the existing defenses for FedRS are insufficient, thus emphasizing the need for better security improvements against our model poisoning attack on FedRS.

D. Recommendation Performance (RQ 3)

Fig. 3 shows the convergence behavior of the proposed HidAttack model in terms of Binary Cross-Entropy (BCE) loss. The training process of the base recommender without any attack (No Attack) and with our proposed attack model (HidAttack) are almost identical, indicating that our attack model does not negatively affect the training process. Additionally, Fig. 3 shows the recommendation performance of the base recommender with and without the attack model in terms of the $NDCG@10$ metric. The results show that our attack model is effective to promote the targeted items while not disturbing the recommendation performance of the base recommender. For the MI-1 M dataset, the $NDCG@10$ metric for the base recommender without any attack is 0.035297, while for the proposed attack model, it is 0.031889, which indicates a small drop in recommendation performance. However, this drop is not statistically significant. This finding suggests that our proposed attack model can effectively

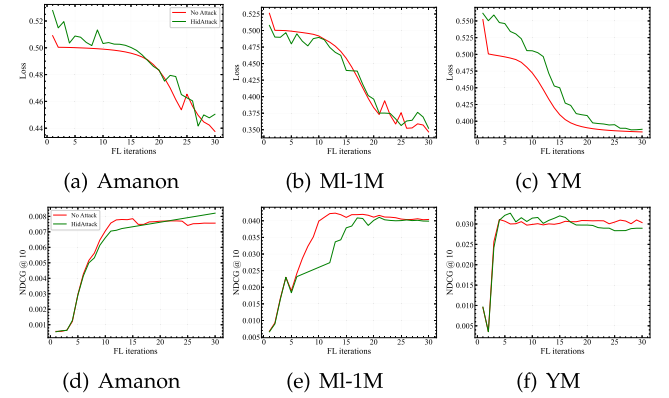


Fig. 3. The convergence behavior of the proposed HidAttack model on three selected datasets. (a), (b), and (c) show the BCE loss for the selected datasets, while (d), (e), and (f) show respective recommendation performance in terms of $NDCG@10$.

promote the targeted items without affecting the recommendation performance of the base recommender.

E. Ablation Analysis

To rigorously evaluate the impact of HidAttack, we performed an ablation analysis by comparing it against a baseline ensemble model. This ensemble model was constructed by simply stacking the four baseline attack models $\mathcal{A}_1$ through $\mathcal{A}_4$ without any bandit support. Unlike HidAttack that leverages an intelligent bandit approach, the ensemble model passively combines the outputs of the baseline attacks. In this ensemble setup, we introduced an additional layer that aggregates the maximum poisoning impact from each baseline model, subsequently calculating the exposure ratio $ER@5$ for the resulting attack. In contrast, HidAttack dynamically adjusts the poisoned gradients, leveraging the exploration-exploitation trade-off to enhance the exposure ratio of targeted items. This configuration allows us to assess the comparative advantage of HidAttack by isolating the influence of the bandit model on attack effectiveness.

Fig. 4 illustrates the comparative performance of the ensemble model versus HidAttack across three datasets, with (a), (b) and (c) depicting trends in $ER@5$, and (d), (e) and (f) showing the corresponding performance in terms of $NDCG@10$. HidAttack consistently achieves higher $ER@5$ values, which indicates effective item promotion. Importantly, it has maintained a superior performance across diverse datasets, suggesting robustness of the bandit approach. The recommendation outcomes highlighted in Fig. 4(d)–(f) remain similar between the ensemble model and HidAttack. This indicates that while HidAttack significantly increases the exposure ratio, it does so without compromising the overall recommendation quality.

The ablation analysis clearly highlights the comparative advantage of HidAttack. The bandit model's ability to dynamically adapt and remix poisoned gradients results in a more effective attack, as evidenced by the higher exposure ratios across all datasets. In contrast, the ensemble model's lack of a sophisticated mixing strategy limits its effectiveness, particularly in maintaining a potent and stealthy attack. This analysis confirms that the attack effectiveness is indeed due to the intelligent bandit

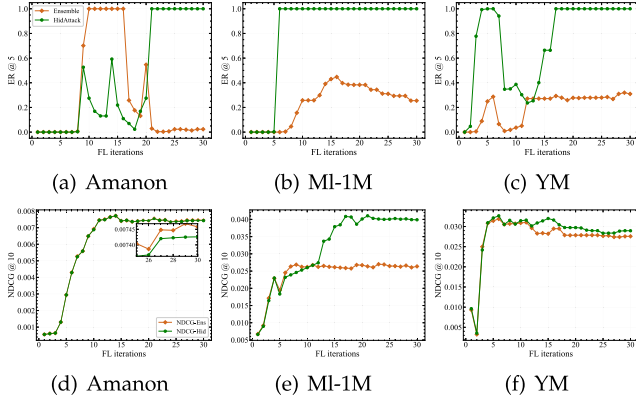


Fig. 4. Comparative analysis of HidAttack and the ensemble model across three datasets. (a), (b), and (c) illustrate the exposure ratio ($ER@5$), demonstrating HidAttack's superior effectiveness due to its intelligent bandit model. (d), (e), and (f) show that the recommendation performance ($NDCG@10$) remains consistent across both models, highlighting HidAttack's ability to enhance attack efficacy without compromising recommendation quality.

model, which successfully balances exploration and exploitation to maximize attack efficacy while preserving recommendation performance.

The theoretical analysis and empirical evaluation conducted in this study reveal a significant vulnerability in the current state of neural collaborative filtering within federated settings, highlighting its susceptibility to sophisticated model poisoning attacks. The proposed attack strategy demonstrates undetectability against state-of-the-art defense mechanisms. Through extensive experimentation with various baseline attacks, the selected attack models ($\mathcal{A}_1$, $\mathcal{A}_2$, $\mathcal{A}_3$, and $\mathcal{A}_4$) have proven to be the most effective for this purpose. The primary goal of this work is to alert industry professionals to the potential risks associated with deploying NCF in federated systems, emphasizing the necessity for heightened security measures.

A key limitation identified during this study pertains to the management of malicious user proportions. Although the proportion of malicious clients has been significantly reduced from the 10-40% range reported in previous works [10], [14] to 1-5%, managing even 1% of malicious clients in large-scale systems still remains a practical challenge. Future research directions should focus on two primary areas: (1) optimizing the given attack strategy to function effectively with an even smaller set of malicious users, and (2) evaluating the proposed attack model on other foundational recommender backbones, such as factorization machines and LightGCN. Furthermore, it is crucial to explore the development of adaptive defense mechanisms capable of countering evolving attacks like HidAttack, thereby enhancing the resilience of FedRS against such sophisticated threats. These advancements will contribute to a more comprehensive understanding of both attack vulnerabilities and defense strategies in FedRS.

VI. CONCLUSION

In this paper, we present an effective and hidden model poisoning attack (HidAttack) to federated recommender systems for targeted item promotion. We address three limitations in

existing attack models: (1) attack with a limited number of Byzantine clients, (2) stealth attack behavior, i.e., bypassing defense mechanism at FL server, and (3) attack execution without knowing side information of other benign users in the system. The poison generation behavior of the proposed attack model is dynamic and therefore undetectable to defensive protocols. We stack a layer of attack models that generate poisoning gradients and on top of this attack cluster, a bandit approach is used that intelligently remixes poison impact such that the poisoned gradients bypass the defense mechanism at the FL server. The experimental evaluation highlights the superiority of the proposed HidAttack over peer attack methods even in the presence of recent defense strategies. Our findings reveal the fact that the existing defenses for FedRS are insufficient, thus emphasizing the need for better security updates in the current state of FedRS for fair recommendation services.

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